Prepared by Audrey Rah (Student 6)
Integration & Dashboard Lead
Group 01 – Multimodal Crime / Incident Report Analyzer
EDS 6397- Generative AI
University of Houston, Summer 2026
Professor: Dr. Rajan Karthik

Here are the links:
Incident Tracker Dashboard1: https://incident-tracker.audreyrah.com/
Data Pipeline Verification Dashboard2: https://pipeline.audreyrah.com/]

Repository Link (Dashboard Folder)

Dashboard Repository:
https://github.com/Audrey2023uh/EDS6397-Group01-Incident-Analysis/tree/main/dashboard

Overview

The integration and dashboard component was developed as the final stage of the EDS6397 Group 01 Multimodal Incident Report Analyzer project. After each team member completed modality-specific processing for Audio, Document, Image, Video, and Text, the integration lead received all five structured CSV outputs, validated them against a shared schema, merged them into a single unified dataset, and built two web-based dashboards for operational monitoring and pipeline verification. The integration workflow followed the professor's five-step assignment structure: (1) assign modality-prefixed Incident_ID values, (2) concatenate all five DataFrames using a UNION merge, (3) handle missing values with the literal text N/A, (4) generate a derived Severity field from Confidence_Score, and (5) build a dashboard interface for filtering and investigating incident summaries. The final deliverables include the integrated dataset (dashboard/data/dataset.csv), Dashboard 1 (Real-Time Incident Tracker), Dashboard 2 (Data Pipeline Verification Dashboard), a workflow and architecture diagram, and supporting documentation in the project GitHub repository. This component demonstrates that outputs from five independent unstructured-data pipelines can be standardized, validated, merged, and presented through a unified operational interface suitable for emergency incident review and integration transparency.



Input Datasets

The integration component received structured CSV outputs from all five modality teams. Each file was produced by a separate student pipeline and was required to conform to the team's seven-column common schema before merge. The standardized input files used during integration were: The five modality input files, record counts, and original Incident_ID ranges are summarized in Table 1.

Table 1. Input Dataset Summary by Modality

Input File	Source Type	Records	Original Incident_ID Range
audio.csv	Audio	708	INC-2026-000001 – INC-2026-000708
pdf.csv	Document	13	INC-2026-001000 – INC-2026-001012
image.csv	Image	72	INC-2026-002000 – INC-2026-002071
video.csv	Video	344	INC-2026-003000 – INC-2026-003343
text.csv	Text	115	INC-2026-004000 – INC-2026-004114

Each modality dataset was generated independently. The Audio component processed 911 emergency call audio; the Document component processed Arkansas 1033-program PDF records; the Image component processed scene-detection outputs; the Video

component processed CAVIAR CCTV clips; and the Text component processed CrimeReport social/news text records. Because the modalities do not describe the same real-world incidents, integration was performed as a schema-aligned UNION rather than a relational join. Before merge, each input file was reviewed to confirm the presence of all required columns, correct Source_Type values, valid Confidence_Score ranges, and non-overlapping Incident_ID blocks as defined in the team repository.

Integration Tools and Libraries

The integration and dashboard work was implemented using the following tools and libraries:

Table 2 lists the tools and libraries employed in the integration and dashboard work.

Table 2. Integration Tools and Libraries

Tool / Library	Role in Integration and Dashboard Work
Python	Dataset validation, merge logic, dashboard data embedding, and metric computation
Pandas	Loading modality CSV files, schema alignment, and pandas.concat UNION merge
NumPy	Supporting numeric operations during validation and summary statistics
HTML / CSS / JavaScript	Dashboard user interfaces for App 1 and App 2
Plotly	Interactive charts in Dashboard 1 (event distribution, source breakdown, hotspots, timeline)
Matplotlib	Supporting figures for cross-modality integration analysis in the incident_response folder

Dashboard data was embedded directly into each index.html file using Python scripts (embed_dataset.py for Dashboard 1 and embed_data.py for Dashboard 2). This design allows both dashboards to run locally in a browser without a backend server and without external file-loading restrictions.

Integration Methodology

The integration process was executed in the order defined by the course assignment and documented in dashboard/data/dataset_preparation.md. The methodology consisted of six major stages: schema validation, Incident_ID assignment, UNION concatenation, missing-value handling, severity generation, and final dataset validation. After the integrated dataset was created, two dashboards were built to support operational review and backend verification. The guiding principle throughout integration was preservation of one record per source output row. No join was performed on Incident_ID, because incidents from different modalities are independent records in this prototype. The integration task therefore focused on schema compatibility, identifier uniqueness, and consistent derived-field generation rather than cross-source incident resolution.

Schema Standardization

All five modality outputs were standardized to the team common schema before concatenation. The required columns were:

The seven required columns standardized across all modality outputs are defined in Table 3.

Table 3. Required Common Schema Columns

Column	Description
Incident_ID	Unique identifier in team format before prefix assignment
Event_Type	Detected or labeled event category
Source_Type	Audio, Document, Image, Text, or Video
Date_Time	Timestamp string or N/A
Location	Location string or N/A
Confidence_Score	Numeric confidence value between 0.0 and 1.0
Summary	Human-readable incident summary

During integration, two additional columns were added to the final dataset:

Table 4 describes the two additional columns generated during integration.

Table 4. Integration-Derived Columns Added to the Final Dataset

Added Column	Description
Severity	Derived Low / Medium / High classification from confidence
Original_Incident_ID	Pre-prefix team ID preserved for traceability

Schema validation confirmed that all seven required source columns were present in every modality file and that Confidence_Score values fell within the valid range [0, 1] across all 1,252 final records.

Incident_ID Assignment

To prevent duplicate identifiers across modalities, the team adopted reserved Incident_ID blocks at the project level. During integration Step 1, each row received a modality prefix while the original team ID was preserved in Original_Incident_ID. Modality-prefixed Incident_ID assignment rules and record counts are presented in Table 5.

Table 5. Modality-Prefixed Incident_ID Assignment

Source Type	Prefix	Example Integrated ID	Record Count
Audio	AUD-	AUD-000001	708
Document	DOC-	DOC-001000	13
Image	IMG-	IMG-002000	72
Video	VID-	VID-003000	344
Text	TXT-	TXT-004000	115

Validation confirmed that all 1,252 integrated IDs were unique, that only the five expected prefixes were present (AUD, DOC, IMG, VID, TXT), and that each Source_Type mapped to the correct prefix family. No duplicate Incident_ID values were found in the final dataset.

Dataset Concatenation (UNION)

After schema validation and prefix assignment, the five standardized DataFrames were combined using pandas.concat as a UNION operation. This produced one integrated table with one row per source record and no relational join across modalities.

Table 6 reports the number of records merged from each modality through the UNION concatenation.

Table 6. UNION Merge Record Counts by Modality

Modality	Rows Merged
Audio	708
Document	13
Image	72
Video	344
Text	115
Total	1,252

The UNION approach was appropriate because each modality describes separate prototype incident records. Concatenation preserved all source rows while aligning them to a single common schema for dashboard filtering, validation, and reporting. The final merged output was saved as dashboard/data/dataset.csv.

Missing Value Handling

Missing or unavailable field values were handled using the literal text N/A, as required by the assignment. This rule was applied only where a modality could not supply a value for a given column.

Table 7 documents the missing-value handling policy and final N/A counts by column.

Table 7. Missing Value Handling by Column

Column	Missing-Value Policy	Result in Final Dataset
Incident_ID	Must not be missing	0 missing
Event_Type	Must not be missing	0 missing
Source_Type	Must not be missing	0 missing
Confidence_Score	Must not be missing	0 missing
Summary	Must not be missing	0 missing
Original_Incident_ID	Must not be missing	0 missing
Date_Time	Fill with N/A when unavailable	72 missing values
Location	Fill with N/A when unavailable	72 missing values

All 72 N/A values in Date_Time and Location belong to the Image modality, which did not provide timestamp or geographic location fields in its source output. No other required column contained missing values after integration. Across all required fields, the dataset achieved 98.4% cell completeness (8,620 of 8,764 required cells populated).

Severity Generation

A derived Severity column was added during integration Step 4 using the professor's confidence-based rule. Severity was not supplied by the individual modality teams; it was computed uniformly for all 1,252 records after merge.

Formula:

severity_score = Confidence_Score × 10

Classification thresholds:

The confidence-based severity thresholds applied in Step 4 are specified in Table 8.

Table 8. Severity Classification Thresholds

Severity	Threshold (severity_score)
Low	$0 \leq \text{score} < 3$
Medium	$3 \leq \text{score} < 7$
High	$7 \leq \text{score} \leq 10$

Final severity distribution:

The resulting severity distribution in the final dataset is shown in Table 9.

Table 9. Final Severity Distribution in dataset.csv

Severity	Count
High	609
Medium	407
Low	236

Validation confirmed that every row in dataset.csv matched the threshold rule exactly. The final implementation uses only three severity tiers: Low, Medium, and High. No Critical tier appears in the integrated dataset or either dashboard.

Limitations and Assumptions

Following the project requirements, Severity was generated during the integration stage using each record's Confidence_Score to provide a consistent classification across all modalities. Likewise, the Actionable field was derived from the Event_Type using rule-based logic to support dashboard filtering and incident prioritization. This unified integration approach provides consistency across heterogeneous pipelines; however, Confidence_Score reflects the confidence of the source pipeline rather than the true operational priority of an incident. Consequently, a record with a high confidence score is not necessarily a high-priority emergency, while a lower-confidence record could still represent a critical incident. For example, a document may be extracted with very high confidence while describing a low-priority event. Therefore, the Severity and Actionable fields in this prototype should be interpreted as integration-level indicators rather than definitive operational risk assessments. In a production system, these fields would ideally incorporate modality-specific scoring models together with additional contextual information, such as incident type, threat level, operational policies, historical context, and other domain-specific factors.

Data Validation

Before and after integration, the dataset was validated for schema consistency, identifier integrity, missing-value handling, confidence range compliance, and severity-rule correctness. Dashboard 2 automates much of this validation at build time through embed_data.py, which reads dataset.csv and computes integration metrics for display. Validation results from the final integrated dataset:

Table 10 summarizes the integration validation results from the final dataset.

Table 10. Integration Validation Results

Validation Check	Result	Detail
Schema Validation	PASS	All 7 required columns present (+ Severity, Original_Incident_ID)
Source Integration	PASS	5 source types integrated: Audio, Document, Image, Text, Video
Missing Values	WARN	144 N/A values across required columns (72 Date_Time + 72 Location)
Duplicate Records	PASS	0 duplicate Incident_ID values
Incident_ID Prefix	PASS	Modality prefix format met for AUD, DOC, IMG, VID, TXT
Date Parsing	WARN	1,043 of 1,252 Date_Time values parseable (83.3%)
Confidence Range	PASS	All Confidence_Score values in [0, 1]
Severity Rules	PASS	Low / Medium / High assigned from Confidence_Score × 10
Data Completeness	PASS	98.4% cell completeness across required columns

The integrated dataset contains 57 distinct event types, mean confidence of 0.572, median confidence of 0.69, and 921 records classified as Actionable under the dashboard rule described in the Dashboard 1 section below.

Integration Pipeline Architecture

The end-to-end integration architecture is documented in the workflow diagram titled Multimodal Incident Dataset Pipeline & Dashboards: From Five Modalities to Actionable Incident Intelligence, stored in the repository under dashboard/Multimodal Incident Dataset Pipeline Diagram/. The diagram summarizes the full pipeline from five modality CSV inputs through validation, merge, severity generation, and dashboard delivery.

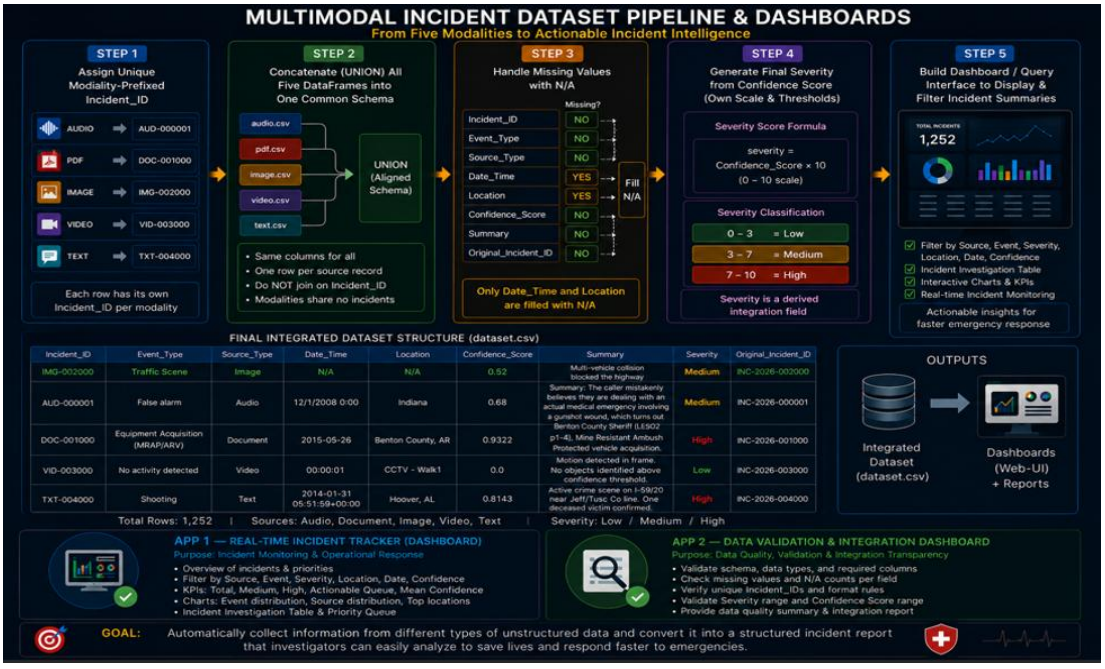


Figure 1. Multimodal Incident Dataset Integration Pipeline and Dashboard Architecture

Figure 1 illustrates the complete integration workflow developed for this project. The architecture begins with the five modality-specific datasets, performs schema validation, Incident_ID assignment, dataset concatenation, missing-value handling, and confidence-based severity generation, and concludes with the development of the Real-Time Incident Tracker and the Data Pipeline Verification Dashboard. The figure summarizes the end-to-end integration process that transforms independent modality outputs into a unified incident dataset for analysis and operational monitoring. The architecture follows five sequential stages:

The architecture follows five sequential stages:

1. Assign modality-prefixed Incident_ID values to each source row.

2. Concatenate (UNION) all five DataFrames into one aligned schema.

3. Fill missing Date_Time and Location values with N/A.

4. Generate Severity from Confidence_Score × 10.

5. Build dashboards for operational monitoring and integration verification.

The pipeline outputs are:

- dashboard/data/dataset.csv — final integrated dataset
- Dashboard 1 — Real-Time Incident Tracker
- Dashboard 2 — Data Pipeline Verification Dashboard
- Integration analysis figures in dashboard/incident_response/

This architecture separates data engineering (Steps 1–4) from presentation and validation (Step 5), allowing the integrated dataset to be reviewed independently before operational use.

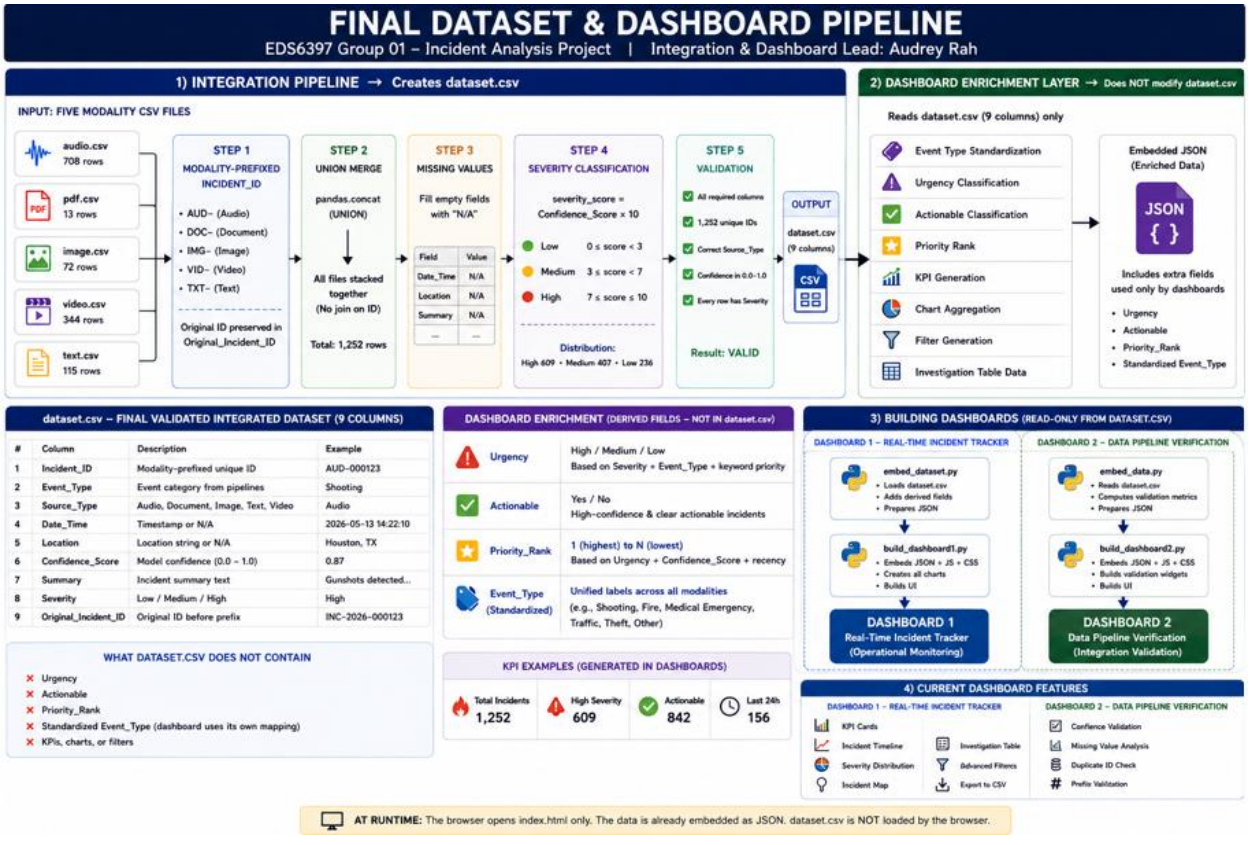


Figure 2. Integration Pipeline and Dual Dashboard Architecture

Figure 2 presents the complete workflow from multimodal data integration to the development of the two dashboards. After schema validation, modality-prefixed Incident_ID assignment, UNION dataset concatenation, missing-value handling, and confidence-based severity classification, the validated integrated dataset is used as the single source of truth for two independent applications: Dashboard 1 (Real-Time Incident Tracker) for operational incident monitoring and Dashboard 2 (Data Pipeline Verification Dashboard) for integration validation and data quality assessment. The figure also illustrates the dashboard build process, derived dashboard fields, and the separation between the validated dataset and dashboard-specific enrichment.

Final Integrated Dataset

The final integrated dataset is stored at dashboard/data/dataset.csv. It contains 1,252 records and 9 columns.

Final schema:

The complete final integrated dataset schema is provided in Table 11.

Table 11. Final Integrated Dataset Schema

Column	Description
Incident_ID	Modality-prefixed unique ID (e.g., AUD-000001)
Event_Type	Event label assigned by the source modality
Source_Type	Audio, Document, Image, Text, or Video
Date_Time	Timestamp string or N/A
Location	Location string or N/A
Confidence_Score	Numeric confidence between 0.0 and 1.0
Summary	Incident summary text
Severity	Low, Medium, or High
Original_Incident_ID	Original team ID before prefixing

Representative records from the final dataset:

Table 12 presents representative records drawn from the final integrated dataset.

Table 12. Representative Records from the Final Integrated Dataset

Incident_ID	Event_Type	Source_Type	Date_Time	Location	Confidence_Score	Severity	Original_Incident_ID
IMG-002000	Traffic Scene	Image	N/A	N/A	0.52	Medium	INC-2026-002000
AUD-000001	False alarm	Audio	12/1/2008 0:00	Indiana	0.68	Medium	INC-2026-000001
DOC-001000	Equipment Acquisition (MRAP/ARV)	Document	2015-05-26	Benton County, AR	0.9322	High	INC-2026-001000
VID-003000	No activity detected	Video	00:00:01	CCTV - Walk1	0.0	Low	INC-2026-003000
TXT-004000	Shooting	Text	2014-01-31 05:51:59+00:00	Hoover, AL	0.8143	High	INC-2026-004000

This dataset is the single source of truth for both dashboards, the workflow diagram, integration documentation, and the final project analysis.

Dashboard 1 – Real-Time Incident Tracker

Dashboard 1 is the user-facing emergency operations interface. It allows investigators and operators to filter, prioritize, and review incident summaries from the integrated dataset in a single web view. The dashboard is implemented as a self-contained HTML application with embedded data and Plotly charts, located in dashboard/dashboard1_real_time_incident_tracker/.

Primary files:

The primary files comprising Dashboard 1 are listed in Table 13.

Table 13. Dashboard 1 Primary Repository Files

File	Purpose
index.html	Main dashboard with embedded const DATA = {...}
embed_dataset.py	Rebuilds embedded data from dataset.csv
README.md	Dashboard documentation
app1.png	Screenshot for report and repository documentation

Operational features:

- Filter by Source Type, Event Type, Severity, Location, minimum confidence, and text search
- Incident Priority Queue showing top actionable High-severity incidents
- Interactive charts for event-type distribution, source-type breakdown, location hotspots, and incident timeline
- Key Data Insights panel with severity summaries and top locations/event types
- Incident Investigation Table with full record detail and severity badges
- Live clock and dataset metadata banner

KPI definitions (default unfiltered counts):

Default unfiltered KPI counts displayed in Dashboard 1 are defined in Table 14.

Table 14. Dashboard 1 Key Performance Indicators

KPI	Definition	Count
Total Incidents	Records in the filtered set	1,252
Medium Priority	Severity = Medium	407
High Priority	Severity = High	609
Actionable Queue	Actionable = true AND Severity = High	578
Mean Confidence	Mean Confidence_Score	0.57

Actionable rule (dashboard-derived, not stored in CSV):

Actionable = false when Event_Type is blank, N/A, Unknown, False alarm, No activity detected, or Person walking / present; otherwise Actionable = true. A total of 921 records are Actionable under this rule.

Dashboard 1 implements assignment Step 5 by converting the integrated dataset into an operational incident-monitoring interface for rapid situational awareness and review.

Dashboard 2 – Data Pipeline Verification Dashboard

Dashboard 2 is the backend integration and validation interface. Its purpose is to verify that the multimodal merge pipeline produced a schema-consistent, identifier-safe, and analytically usable dataset before that dataset is used operationally in Dashboard 1. It is implemented as an HTML/CSS/JavaScript application with metrics computed at build time from dataset.csv.

Primary files:

The primary files comprising Dashboard 2 are listed in Table 15.

Table 15. Dashboard 2 Primary Repository Files

File	Purpose
index.html	Main dashboard with embedded const DASHBOARD_DATA = {...}
embed_data.py	Computes validation metrics and embeds them into index.html
script.js	Client-side rendering logic
styles.css	Dashboard styling
README.md	Dashboard documentation
app2.png	Screenshot for report and repository documentation

Verification features:

- Schema validation against the seven required columns
- Source integration summary across all five modalities
- Missing-value analysis by column
- Duplicate Incident_ID check
- Modality prefix verification (AUD, DOC, IMG, VID, TXT)
- Date parsing coverage report
- Confidence range and distribution analysis
- Severity rule verification
- Overall data quality summary

Dashboard 2 functions as the integration validation pipeline in visual form. It documents whether the merged dataset is structurally sound, complete enough for analysis, and consistent with the professor's integration rules.

Results

The integration and dashboard component successfully completed all assigned project objectives.

Integration results:

- Received and validated outputs from all five modalities
- Merged 1,252 records into one unified dataset using pandas.concat
- Assigned modality-prefixed Incident_ID values with zero duplicates
- Filled 144 missing Date_Time/Location cells with N/A
- Generated Severity for all records using the confidence-based rule
- Produced the final dataset at dashboard/data/dataset.csv

Dashboard results:

- Built Dashboard 1 for real-time incident monitoring, filtering, prioritization, and investigation
- Built Dashboard 2 for schema, identifier, missing-value, and severity validation
- Deployed both dashboards as live web applications
- Produced workflow/architecture documentation and repository README files
- Generated cross-modality integration figures for the incident_response folder

Key summary statistics:

Key integration and dashboard summary statistics are consolidated in Table 16.

Table 16. Key Integration and Dashboard Summary Statistics

Metric	Value
Total integrated records	1,252
Source types	5
Event types	57
High severity records	609
Medium severity records	407
Low severity records	236
Actionable records	921
Actionable High-severity queue	578
Mean confidence	0.572
Parseable dates	1,043 (83.3%)
Required-column completeness	98.4%

The integration component demonstrated that multimodal incident outputs can be standardized and merged into one analyzable dataset and then presented through both an operational dashboard and a validation dashboard within the same project repository.

Challenges and Solutions

Several challenges arose during integration and dashboard development.

Schema consistency across modalities. The five source pipelines used different upstream field names and formats before team standardization. This was addressed by validating each modality CSV against the seven-column common schema before merge and rejecting any file that did not conform.

Highly uneven dataset sizes. Audio contributed 708 records while Document contributed only 13. UNION concatenation preserved all rows without forcing equal representation, allowing the dashboards to reflect true modality volume rather than artificial balancing.

Missing Image timestamps and locations. All 72 Image records lacked Date_Time and Location. These were filled with N/A rather than fabricated values, preserving honesty in the integrated dataset and making the limitation visible in Dashboard 2.

Heterogeneous date formats. Final records used multiple date/time formats, including calendar dates, clip-relative timestamps such as 00:00:01, and timezone-aware text timestamps. This reduced parseable-date coverage to 83.3% and required Dashboard 2 to report date parsing as a warning rather than a hard failure.

Local dashboard deployment. Loading external CSV files directly in a browser can fail under local file:// access rules. Both dashboards solved this by embedding the dataset metrics directly into index.html using Python build scripts, allowing reliable local and hosted use.

Operational prioritization beyond severity. Severity alone does not distinguish false alarms or non-actionable detections from incidents requiring review. Dashboard 1 therefore added an Actionable rule based on Event_Type, producing a more meaningful Actionable Queue of 578 High-severity records.

Limitations and Known Issues

The integration and dashboard implementation has several limitations that should be interpreted in the context of a course prototype rather than a production emergency system. First, the five modality datasets are independent stand-ins. They were not collected from the same real-world incident feed, so the integration demonstrates schema compatibility and pipeline mechanics rather than true cross-source incident correlation. Second, Image records contain no usable Date_Time or Location, which limits timeline and geospatial analysis for that modality. Third, only 83.3% of Date_Time values are parseable into standard timeline analytics because of mixed formats across modalities. Fourth, Actionable is a dashboard-derived heuristic and is not stored in dataset.csv. It improves operational prioritization but is not itself a validated incident label. Fifth, severity is based solely on Confidence_Score and does not incorporate event-type risk tiers beyond what is indirectly reflected in confidence values. Finally, the broad parsed date range in the dataset (from 1010-06-01 to 2104-09-01) indicates that some source timestamps contain noise or formatting anomalies. These values were preserved rather than silently corrected, so downstream users can inspect them through the investigation table and validation dashboard.

Repository and Collaboration

The integration and dashboard work is maintained in the Group 01 GitHub repository EDS6397-Group01-Incident-Analysis. Audrey Rah served as Integration and Dashboard Lead, receiving final modality CSV outputs from the other team members and delivering the unified dataset, dashboards, and integration documentation.

Team roles:

Table 17 identifies the project team members and their assigned roles.

Table 17. Project Team Roles and Responsibilities

Team Member	Role
Kashfin, MD	Audio Processing
Francisco H. Reyes	Document Processing
Alex Landon	Image Processing
Roberto Gomez	Video Processing
Lujia Wu	Text Processing
Audrey Rah	Integration & Dashboard Lead

Repository structure for the integration component:

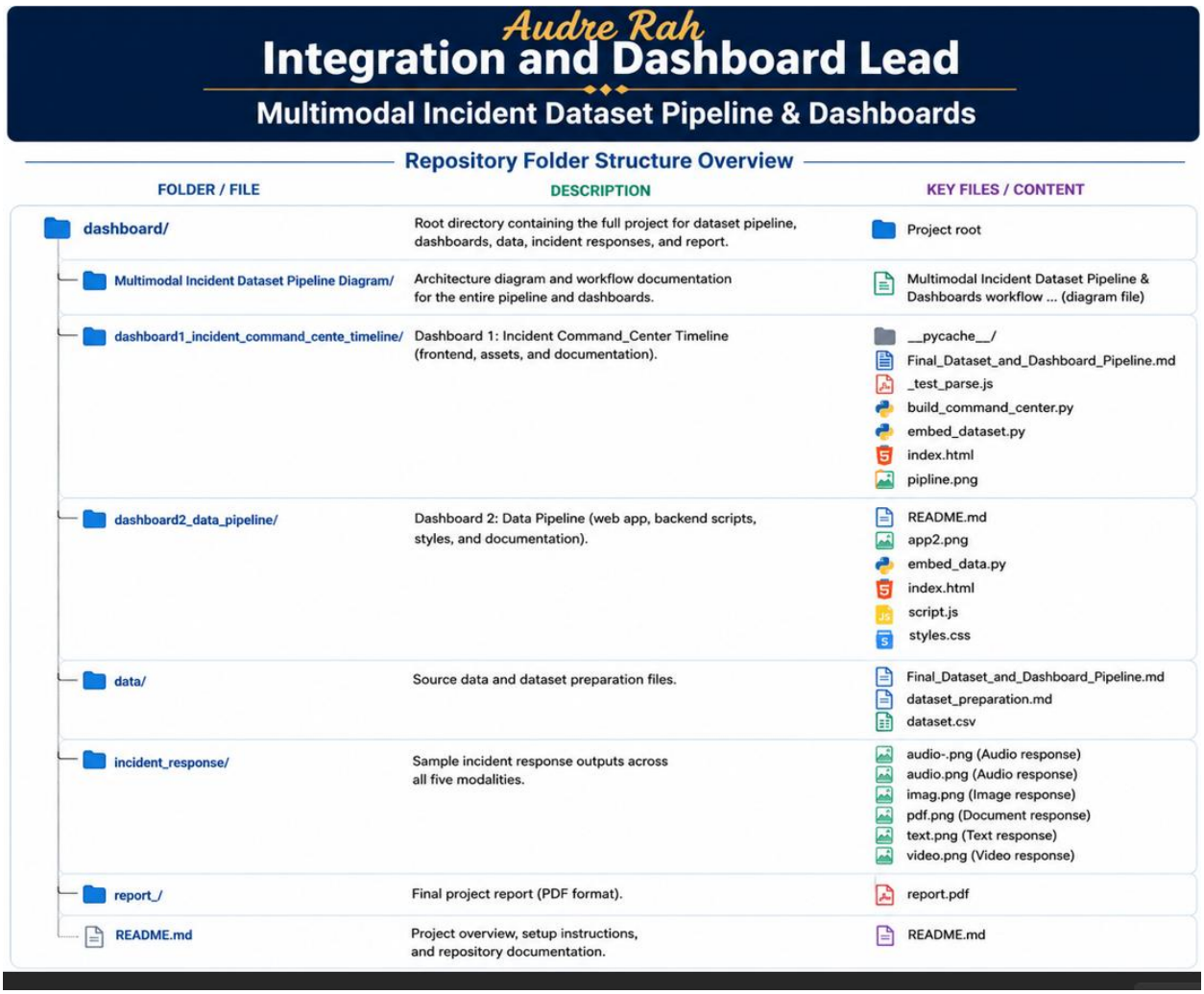


Figure 3. Repository Structure of the Integration and Dashboard Component

Figure 3 illustrates the repository structure of the Integration and Dashboard component, highlighting the organization of the pipeline architecture, dashboards, dataset, incident response outputs, documentation, and supporting project files.

Summary

The integration and dashboard component unified all five modality outputs into a single 1,252-record dataset using schema validation, modality-prefixed Incident_ID assignment, pandas.concat UNION merge, N/A missing-value handling, and

confidence-based severity generation. The resulting dataset was validated for uniqueness, prefix integrity, completeness, and severity-rule compliance, then presented through two complementary dashboards: Dashboard 1 for operational incident monitoring and Dashboard 2 for integration verification. Together, these deliverables complete the final stage of the multimodal incident analysis project by converting separate unstructured-data pipelines into one structured, filterable, and reviewable incident dataset supported by documentation, validation tooling, and live dashboard interfaces. The work fulfills the course integration requirements and provides a reproducible foundation for cross-modality incident analysis within the Group 01 repository.